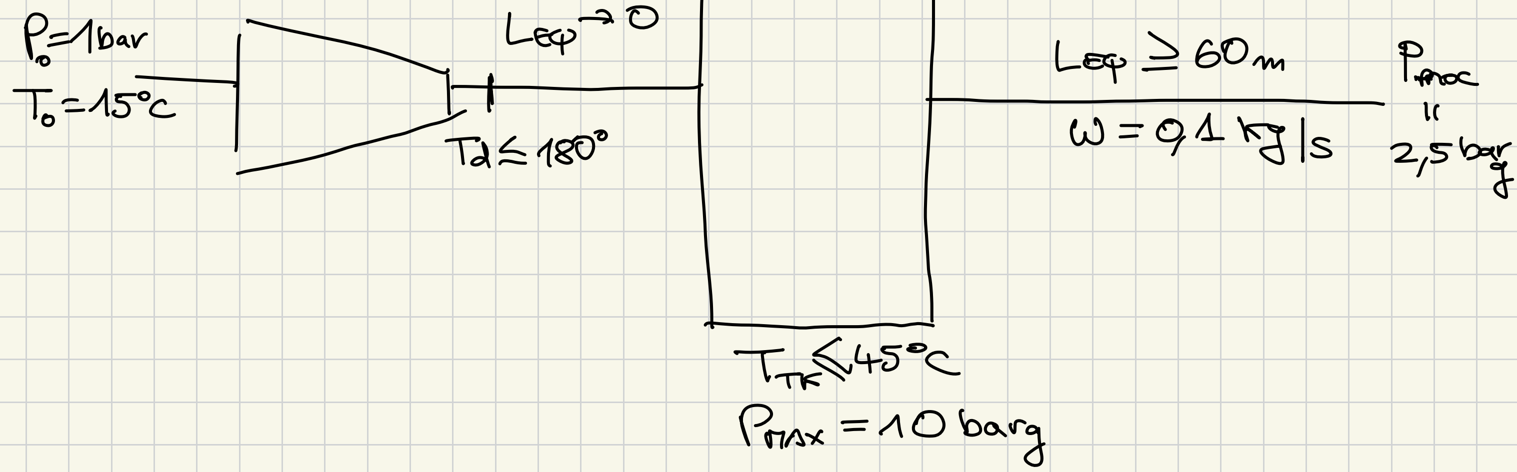
Ej 2



$P_{MAX_{TK}}$ ADMISIBLE:

CONSIDERO $T_d = 180°C$

$\Rightarrow T_{d_{MAX}} = T_0\, r_{c_{MAX}}^{(1-1/k)}$

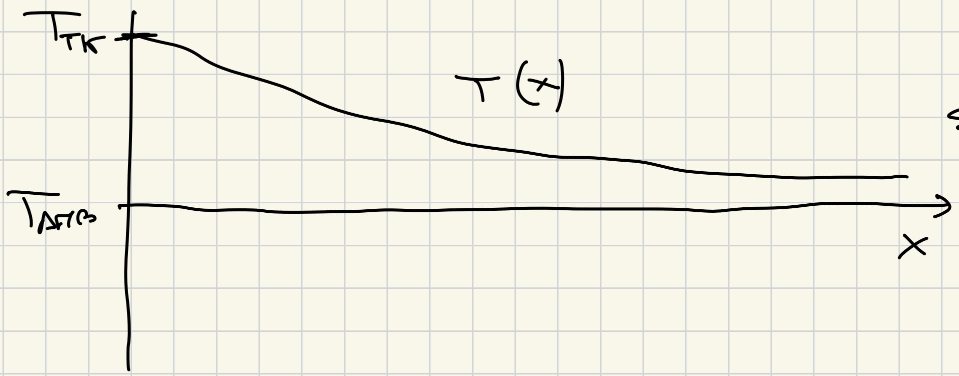
$\Rightarrow r_{c_{MAX}} = 7{,}12 \Rightarrow \boxed{P_{MAX_{TK}} = 7{,}12\,bar}$

$P_{MIN_{TK}}$ QUE ASEGURA w:

① FLUJO COMPRESIBLE ENTRE RESERVORIO Y PROCESO (RESERVORIO $P_3 = cte = 3{,}5\,bar$)

② $T_{TK} \approx 45°C > T_{AMB}$

$\Rightarrow$



SUPONGO $T_2 > T_{AMB}$

HAY QUE CHEQUEARLO AL FINAL

$\Rightarrow w_{ADI} < w_{ISOT}$

$\Rightarrow \Delta P_{ADI} < \Delta P_{ISOT}$

$\Rightarrow (P_1 - P_2)_{ADI} < (P_1 - P_2)_{ISOT}$

$$\Rightarrow \quad P_{1_{ADIA}} - \cancel{P_2} < P_{1_{NORC}} - \cancel{P_2}$$

$$\Rightarrow \quad P_{1_{ADIABÁTICO}} < P_{1_{NORC}}$$

$\Rightarrow$ MODELANDO POR ADIABÁTICO OBTENGO EL MINIMO VALOR DE P_{TA} QUE ASEGURA ω

$\Rightarrow$ ECUACIONES REDUCIDAS, FLUJO ENTRE RESERVORIOS

$$\begin{cases} \left(\dfrac{\gamma-1}{2\gamma}\right) X = r^2 - r^{\gamma+1} & (1) \\ \dfrac{r^2 - z^2}{X} - \dfrac{\gamma+1}{\gamma} \ln\left(\dfrac{r}{z}\right) = \dfrac{4fLe}{D} & (2A')_R \\ \left(\dfrac{\gamma-1}{2\gamma}\right) X = z^2 - Fz & (3A)_R \\ F = P_3/P_0 \end{cases}$$

CON

$$r = \nu_0/\nu_1$$

$$z = \nu_0/\nu_2$$

$$\frac{P_2}{P_0} = F$$

$$X = \left(\frac{\omega}{A}\right)^2 \left(\nu_0/P_0\right)$$

VALORES SEMILLA

$$\begin{cases} r = 0{,}99 \\ X = 0{,}01 \\ \omega = 0{,}1 \leftarrow \text{DATO NO VARÍA} \\ z = 0{,}6 \end{cases}$$

$P_3 = 3{,}5 \times 10^5$

$\Rightarrow P_{0_{MIN}} = 5{,}3 \text{ bar}$